

Chapter 24 — Basic Principles of Immunology and Immunotherapy in Urologic Oncology — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **Th1 (IL-2, TNF- α , IFN- γ)** → **BCG success**; Th2 (IL-4, -5, -10, -13) → BCG failure.
- **BCG PAMPs** → **TLR2/TLR4**; adheres via **fibronectin**; neutrophils hours → CD4 weeks.
- **Renal paradox**: ↑ CD8 density → **worse** RCC outcome.
- **Adaptive immune resistance**: CD8 IFN- γ → induces PD-L1 → switches off CD8.
- **Sipuleucel-T**: OS +4.1 mo; **no PFS effect**; only approved GU vaccine.
- **Nadofaragene**: CR 53.4% CIS; first gene therapy, BCG-unresponsive NMIBC (Dec 2022).
- **First-line pembrolizumab mUC**: cisplatin-ineligible + **CPS ≥ 10** , or platinum-ineligible.
- **KEYNOTE-045**: only randomised phase III basis for anti-PD-1/PD-L1 approval in UC.
- **MSI-H/dMMR** → pembrolizumab; first tissue-agnostic approval (May 2017).
- **CheckMate 214**: nivo + ipi first line int/poor mRCC; OS/ORR won, PFS not significant.

INNATE IMMUNE SYSTEM

- **Neutrophil**: commonest blood leucocyte (~60%); pus.
- **First responder**: tissue-resident macrophage.

CYTOKINES AND CHEMOKINES

- **Prototype chemokine**: CXCL8 (IL-8) → neutrophils.
- **IL-1**: innate cytokine; main fever mediator.
- **IL-12** → **Th1**; **IL-4** → **Th2**; TGF- β → induced Treg.
- **Raised serum IL-2 after BCG** → favourable outcome.

ADAPTIVE IMMUNE SYSTEM

- **DC** = innate-adaptive bridge; homes to node via **CCR7/SLC (CCL21)**.
- **Class I MHC** → **CD8** (~9 aa); **class II** → **CD4**.
- **Treg**: CD4, **FoxP3+**; natural (thymic) population critical.
- **Costimulation**: CD28-CD80/86 = signal 2; TCR-MHC = signal 1.
- **Checkpoint ligands**: PD-1-PD-L1 (B7-H1); CTLA-4-CD80/86; LAG-3-MHC II.

IMMUNE EDITING HYPOTHESIS

- **Immune editing**: elimination → equilibrium → escape.
- **Escape**: ↓ antigen, ↓ MHC, ↑ Tregs.

CHRONIC INFLAMMATION AND THE ENDOGENOUS IMMUNE RESPONSE TO GU CANCERS

- ***S. haematobium***: IARC carcinogen → **squamous** bladder cancer.
- **Other bladder triggers**: chronic UTI, indwelling catheter, cyclophosphamide.
- **Bladder PD-L1**: ↑ with grade; beats WHO grade for prognosis.
- **CD8 infiltration MIBC** → better outcome.
- **Kidney**: weak inflammation link (smoking, obesity).
- **Worst RCC subgroup**: CD8-high + PD-1/LAG-3-high + few APCs.
- **RCC target**: CD38-high macrophages → anti-CD38.
- **Prostate**: >90% tumours peripheral zone = inflamed zone.
- **PIA** (proliferative inflammatory atrophy) adjacent to high-grade PIN.
- **Prostate**: Treg-dominated; tumour cells PD-L1-negative.

IMMUNOTHERAPY FOR GENITOURINARY CANCERS

- **BCG dose**: $1-5 \times 10^8$ viable bacilli; ~50% durable NMIBC remission.
- **IFN- $\alpha 2b$ protein**: active, not durable (1-2 h dwell).
- **Nadofaragene**: adenovirus IFN- $\alpha 2b$ gene + **Syn3**; 3-monthly, 1-h dwell.
- **Nadofaragene**: CR duration 9.69 mo; 24.3% HG-RFS at 12 mo.

- **Vaccine** = adjuvant + antigen; **GM-CSF** recurring adjuvant.
- **Sipuleucel-T**: PAP-GM-CSF (PA2024); IV 2-weekly $\times 3$; 25.8 vs 21.7 mo.
- **Failed vaccines**: IMA901, AGS-003 (RCC); PROSTVAC-VF (prostate).
- **KEYNOTE-057**: pembro BCG-unresponsive CIS; **CR 41%** (Jan 2020).
- **CheckMate 274**: adjuvant nivo MIUC; DFS 20.8 vs 10.8 mo.
- **Adjuvant eligibility**: pT3/pT4a/N+ (no neoadjuvant) or ypT2-4a/ypN+.
- **Maintenance**: avelumab (JAVELIN Bladder 100) → OS gain.
- **Second line**: pembro (OS 10.3 vs 7.4), nivo, avelumab.
- **Withdrawn UC**: atezolizumab (first/second line), durvalumab.
- **CheckMate 025**: nivo > everolimus post-TKI; OS 25.8 vs 19.7.
- **KEYNOTE-564**: adjuvant pembro ccRCC; HR 0.68.
- **KEYNOTE-564 high risk**: pT2 G4/sarcomatoid, $\geq$ pT3, N+, M1 NED.
- **Pembro in mCRPC**: PSA response <10%, ORR <5%.
- **MSI-H/dMMR prostate**: 1.2-12%; ~1/3 Lynch germline.
- **Ipilimumab mCRPC**: PSA response 15-20%; two negative phase III.
- **CheckMate 214**: HR 0.63 OS; ORR 42 vs 27%.
- **IMbassador250**: atezo + enzalutamide, no OS gain.
- **EV**: anti-nectin-4 + MMAE; first ADC mUC (Dec 2019).
- **EV + pembro**: cisplatin-ineligible; ORR 68% (Apr 2023).
- **TKI + ICB first-line mRCC**: all four beat sunitinib.
- **CLEAR** (pembro + lenvatinib): best ORR 71%, PFS 23.9 mo.

CLASSIC TRAPS

- **Trap**: IL-4 is Th1 — **Actually**: IL-4 induces and marks Th2.
- **Trap**: more CD8 always better — **Actually**: worse in RCC.
- **Trap**: PD-L1 constitutive — **Actually**: induced by CD8 IFN- γ .
- **Trap**: Treg is CD8 — **Actually**: CD4, FoxP3+.
- **Trap**: sipuleucel-T delays progression — **Actually**: OS only.
- **Trap**: pembro first line all cis-ineligible — **Actually**: CPS ≥ 10 or platinum-ineligible.
- **Trap**: atezolizumab adjuvant works — **Actually**: IMvigor010 negative; nivolumab positive.
- **Trap**: CheckMate 214 won PFS — **Actually**: P = 0.03 missed 0.009 threshold.
- **Trap**: CTLA-4 mainly on CD8 — **Actually**: on Tregs; PD-1 on CD8.
- **Trap**: PD-1 blockade works in prostate — **Actually**: only MSI-H/dMMR.